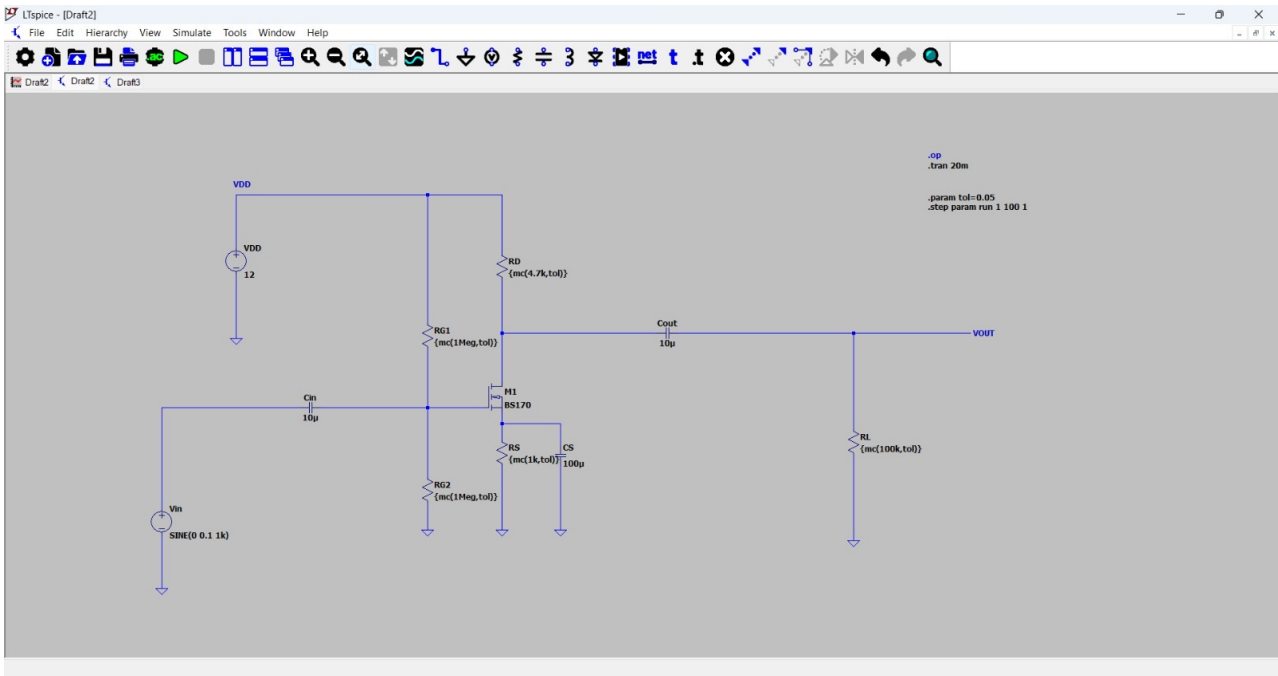


Common-Source NMOS Amplifier Design and Monte Carlo Tolerance Analysis Using LTspice

Portfolio Technical Report

Fortune Ngwenya



A complete analog simulation workflow: bias design, transient response, current analysis, Monte Carlo tolerance modeling, and design interpretation.

Design domain	Analog MOSFET amplifier simulation
Toolchain	LTspice
Device model	BS170 N-channel MOSFET
Supply and input	12 V supply, 100 mV amplitude, 1 kHz sinusoidal input
Verification methods	Transient analysis, current probing, tolerance sweep, Monte Carlo overlay
Tolerance model	+/-5% resistor variation, 100 simulation runs

Executive Summary

This project presents a complete LTspice study of a common-source NMOS amplifier using a BS170 transistor. The design includes a resistive gate-bias network, a drain resistor, source degeneration with bypass capacitance, input/output coupling capacitors, and a load resistor. The simulation workflow begins with a nominal transient verification and progresses into a 100-run Monte Carlo analysis using $\pm 5\%$ resistor tolerance variation.

The goal is not merely to plot a waveform. The purpose is to demonstrate an engineering verification flow: define the circuit, calculate the bias network, run the nominal case, probe meaningful voltages and currents, introduce manufacturing variation, and interpret how tolerance spread affects output response and bias stability.

Primary project outcomes:

- Built and simulated a full common-source NMOS amplifier in LTspice.
- Verified a 1 kHz, 100 mV input stimulus and the resulting drain/output responses.
- Used the drain current through RD to evaluate transistor conduction and operating behavior.
- Implemented a 100-run Monte Carlo simulation with $\pm 5\%$ resistor tolerances.
- Observed output-amplitude and drain-bias variation caused by passive component spread.
- Documented a debugging case involving parameter definition order and corrected the simulation setup.

1. Circuit Design

The amplifier is a single-transistor common-source stage. The input signal is AC-coupled into the gate through C_{in} . R_{G1} and R_{G2} form a DC gate-bias divider. R_D converts transistor drain-current variation into a drain voltage signal. R_S provides source degeneration and DC feedback, while C_S bypasses R_S for AC, increasing small-signal response at the operating frequency. C_{out} blocks the drain DC voltage and couples the AC output to R_L .

Element	Nominal value	Role in circuit
VDD	12 V	DC supply rail
Vin	SINE(0 0.1 1k)	100 mV amplitude, 1 kHz input excitation
M1	BS170	N-channel MOSFET used as the active device
R_{G1} , R_{G2}	1 MOhm each	Gate bias divider, nominal $V_G = 6$ V
R_D	4.7 kOhm	Drain load resistor and current-to-voltage conversion element
R_S	1 kOhm	Source resistor for DC feedback and bias stabilization
C_S	100 uF	Source bypass capacitor for AC gain at 1 kHz
C_{in} , C_{out}	10 uF each	Input and output coupling capacitors
R_L	100 kOhm	Output load

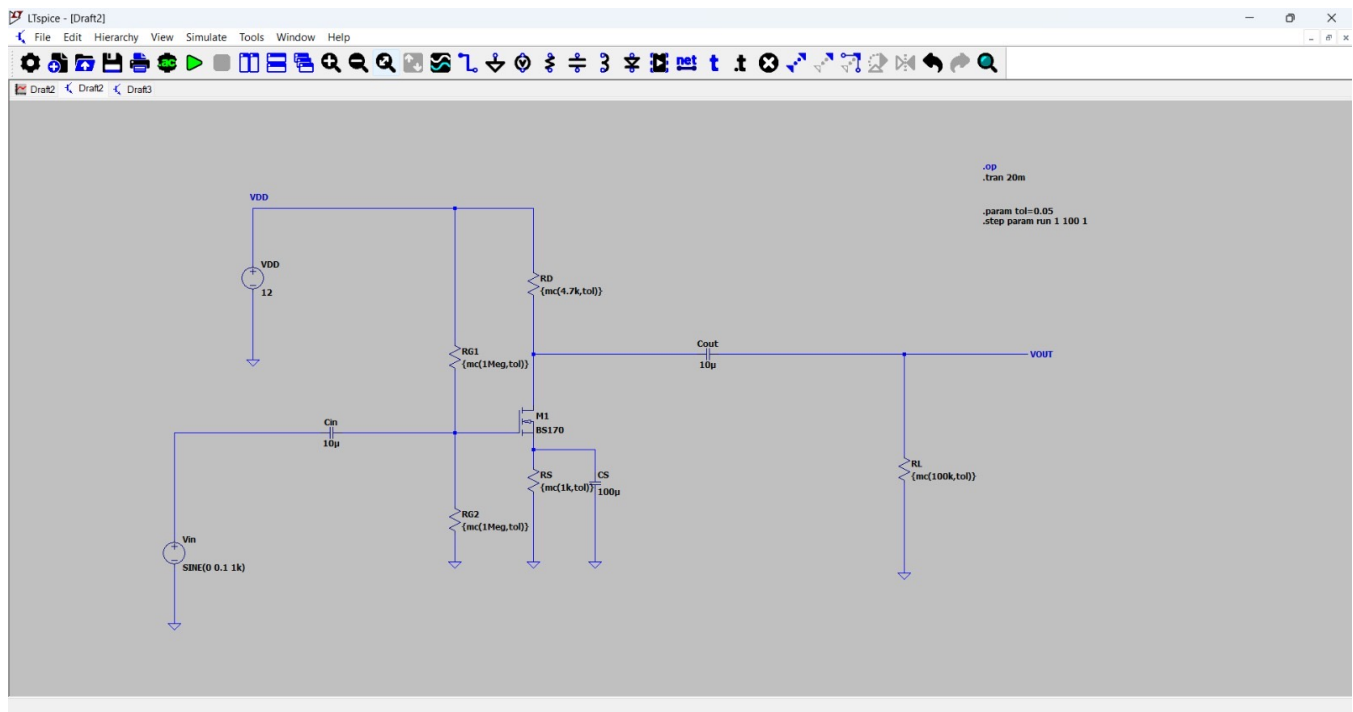


Figure 1. Complete LTspice schematic of the common-source NMOS amplifier with Monte Carlo-capable resistor expressions and the 12 V supply/input excitation defined.

2. Hand Calculations and Bias Reasoning

The first analytical target is the DC gate bias. Because MOSFET gate current is ideally near zero, the gate voltage is set by the resistor divider RG1-RG2.

Nominal gate-bias calculation:

$$V_G = V_{DD} * R_{G2} / (R_{G1} + R_{G2}) = 12 \text{ V} * 1 \text{ MOhm} / (1 \text{ MOhm} + 1 \text{ MOhm}) = 6.0 \text{ V}$$

With both bias resistors given +/-5% tolerance, the approximate extreme gate-voltage range occurs when one divider resistor moves high while the other moves low:

$$\begin{aligned} V_{G,\min} &= 12 * 0.95 / (1.05 + 0.95) = 5.70 \text{ V} \\ V_{G,\max} &= 12 * 1.05 / (0.95 + 1.05) = 6.30 \text{ V} \end{aligned}$$

This predicts a gate bias sensitivity of roughly +/-0.30 V around the 6 V nominal value. That bias movement is significant for a MOSFET stage because drain current depends strongly on VGS.

Coupling and bypass cutoff estimates:

The capacitors were selected so their high-pass corner frequencies are far below the 1 kHz signal. This keeps the capacitors from dominating the response at the test frequency.

Network	Approximate resistance	Capacitance	Cutoff estimate
Input coupling, Cin	RG1 RG2 = 500 kOhm	10 uF	$f_c = 1/(2*\pi*500k*10u) = 0.032 \text{ Hz}$
Output coupling, Cout	RL = 100 kOhm	10 uF	$f_c = 1/(2*\pi*100k*10u) = 0.159 \text{ Hz}$
Source bypass, CS	RS = 1 kOhm	100 uF	$f_c = 1/(2*\pi*1k*100u) = 1.59 \text{ Hz}$

At 1 kHz, all three corner frequencies are far below the operating frequency, so the capacitors behave mainly as AC coupling/bypass elements rather than limiting the test waveform.

Operating-current consistency check:

From the nominal drain plot, the drain node is near 2.14 V. The drain resistor current can be approximated by:

$$I_D \approx (V_{DD} - V_D) / R_D = (12 \text{ V} - 2.14 \text{ V}) / 4.7 \text{ kOhm} \approx 2.10 \text{ mA}$$

This aligns with the LTspice current plots showing approximately 2.10 mA through RD. The source resistor drop at 2.10 mA is approximately 2.10 V, giving VGS near 6.0 V - 2.10 V = 3.90 V.

3. LTspice Simulation Methodology

Three classes of simulation were used: nominal transient analysis, operating-point/debug inspection, and Monte Carlo tolerance analysis.

Mode	Directives	Purpose
Nominal transient	.tran 20m	Used to verify the time-domain response with one deterministic component s
Operating point	.op	Used to inspect steady-state node voltages and catch bias/model issues.
Monte Carlo	.param tol=0.05 .step param run 1 100 1 .tran 20m	Used to run 100 randomized resistor-tolerance cases.

The resistor values were parameterized using LTspice Monte Carlo expressions:

```
RD = {mc(4.7k,tol)}
RS = {mc(1k,tol)}
RG1 = {mc(1Meg,tol)}
RG2 = {mc(1Meg,tol)}
RL = {mc(100k,tol)}
```

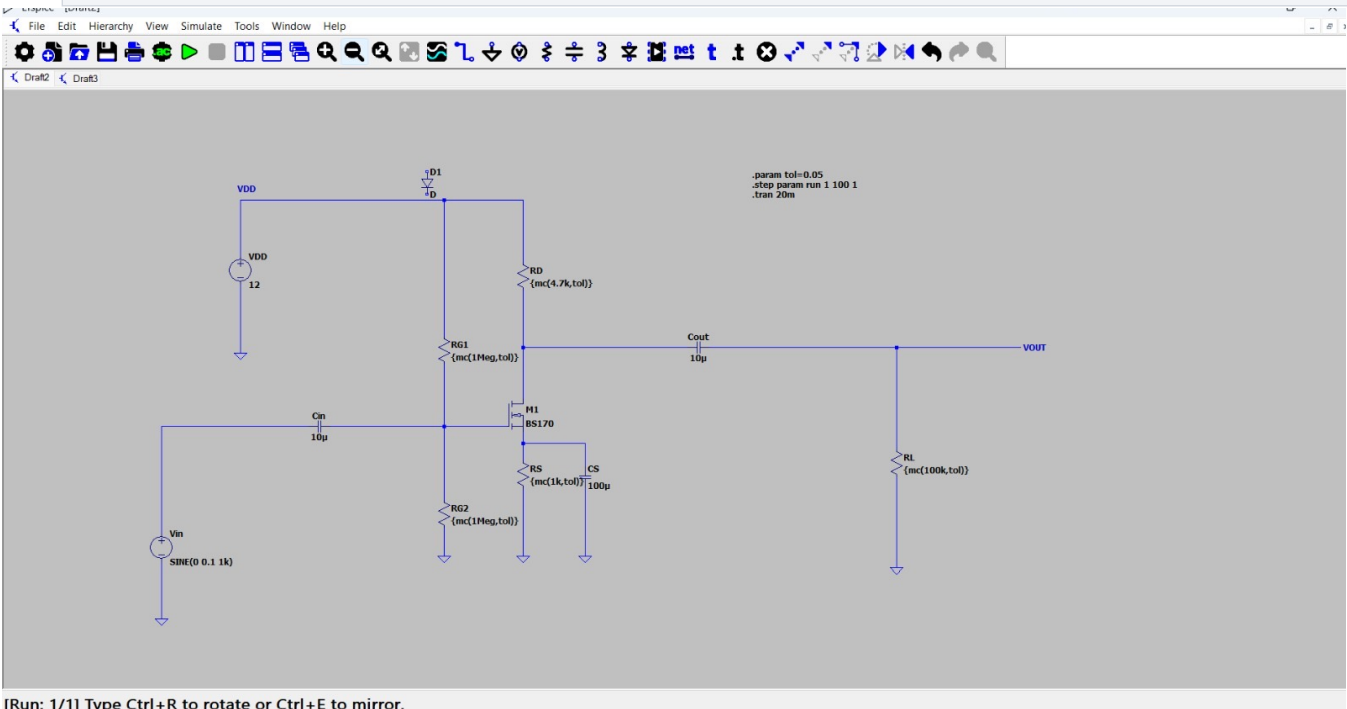


Figure 2. Monte Carlo simulation setup. The schematic contains tolerance-aware component values and simulation commands for a 100-run sweep.

4. Nominal Transient Results

The input source is a 1 kHz sine wave with 100 mV amplitude. This stimulus establishes the small-signal test condition for the amplifier.

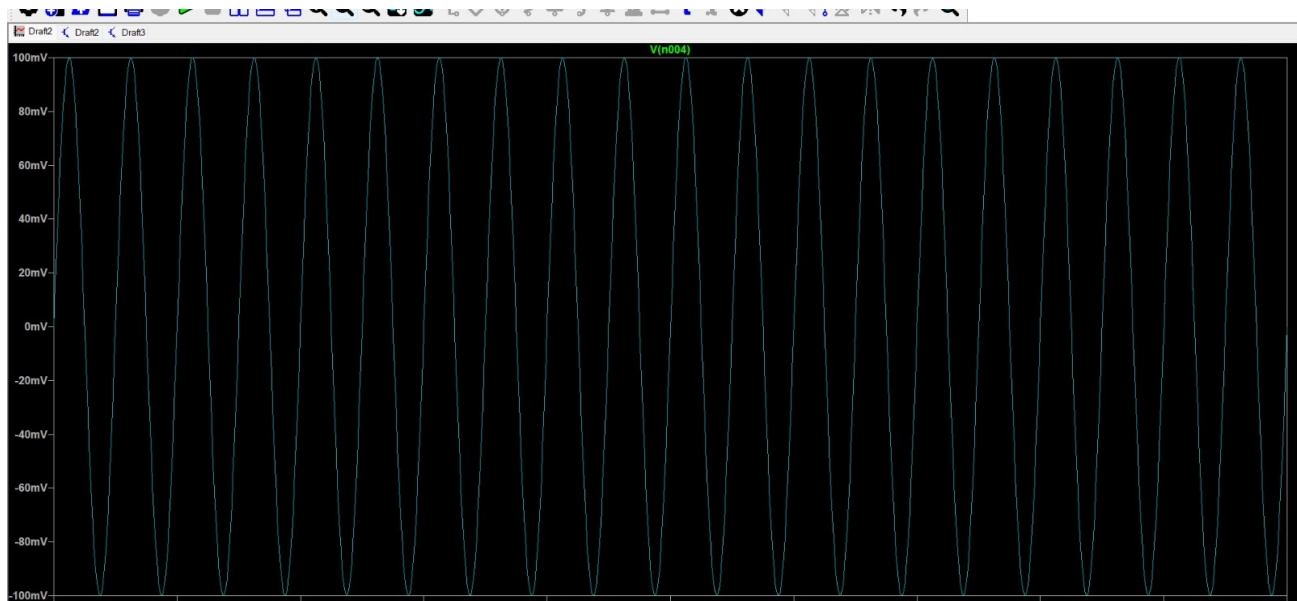


Figure 3. Input excitation waveform: 100 mV amplitude, 1 kHz sinusoid.

The drain node response is measured before the output coupling capacitor. This node includes the DC bias component plus the small AC movement created by drain-current modulation.

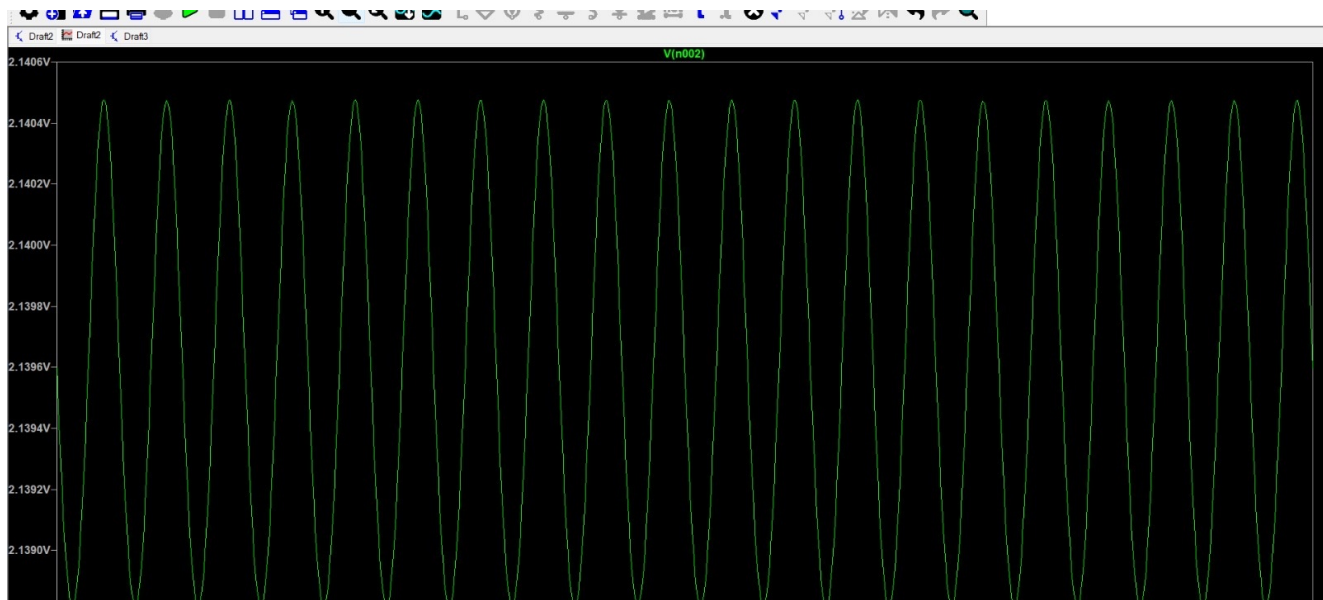


Figure 4. Nominal drain-node voltage response. The drain voltage sits near the DC operating point and contains a small sinusoidal variation.

5. Node and Current Interpretation

The combined waveform view shows multiple circuit quantities plotted together. This type of view is useful for checking consistency between voltage movement at the drain/output nodes and current through RD.

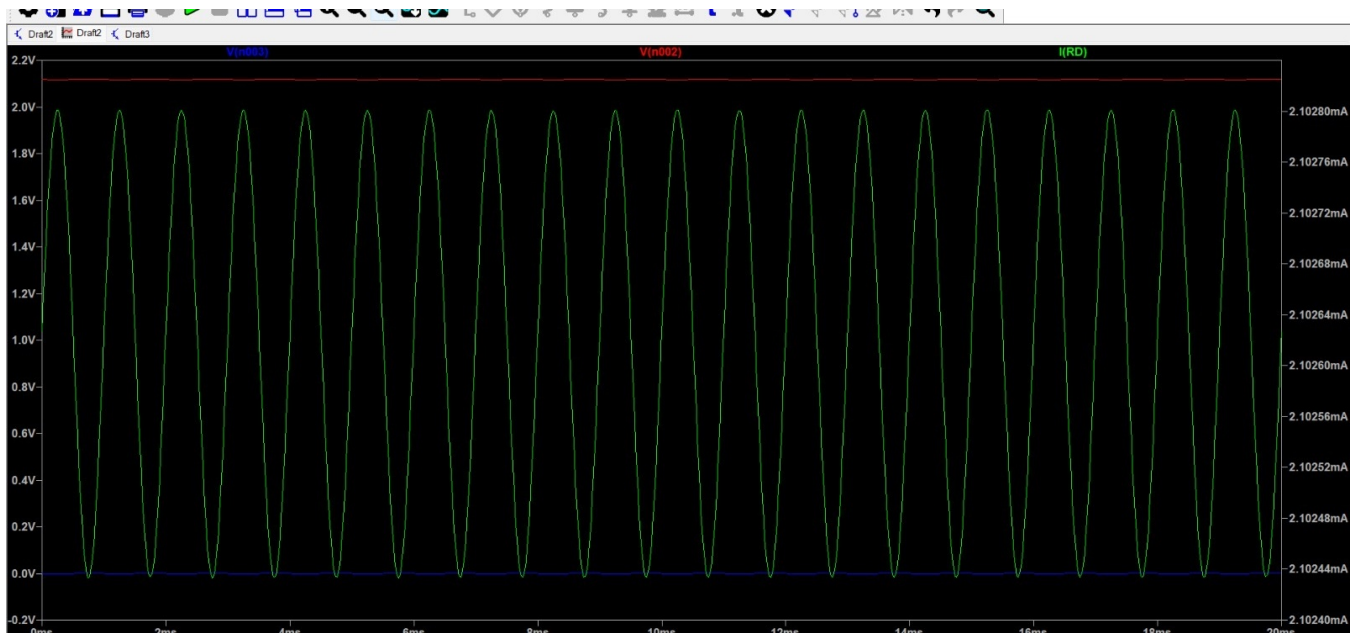


Figure 5. Combined nominal waveform view including drain/output-related voltages and RD current. The current through RD is the key current path for the MOSFET stage.

The drain current estimated from the plot is approximately 2.10 mA. This is consistent with the DC voltage drop across RD and confirms that the transistor is conducting. The observed output amplitude is small because the selected BS170 model and bias point place the transistor in a strongly conducting low-VDS region. This is an important engineering result: simulation revealed not only that the circuit operates, but also that its bias could be further optimized for higher voltage gain by shifting the drain DC level closer to mid-supply.

Design insight:

A higher-gain version would typically adjust RG1/RG2, RD, and RS to bias the drain closer to about VDD/2. The present project focuses on building the verification workflow and quantifying tolerance sensitivity, while also identifying this bias optimization path.

6. Monte Carlo Tolerance Analysis

Monte Carlo simulation was used to evaluate how passive component tolerances affect circuit behavior. Each run randomly perturbs the resistor values by $\pm 5\%$. Across 100 runs, the output waveform spreads into a family of possible responses instead of a single deterministic curve.

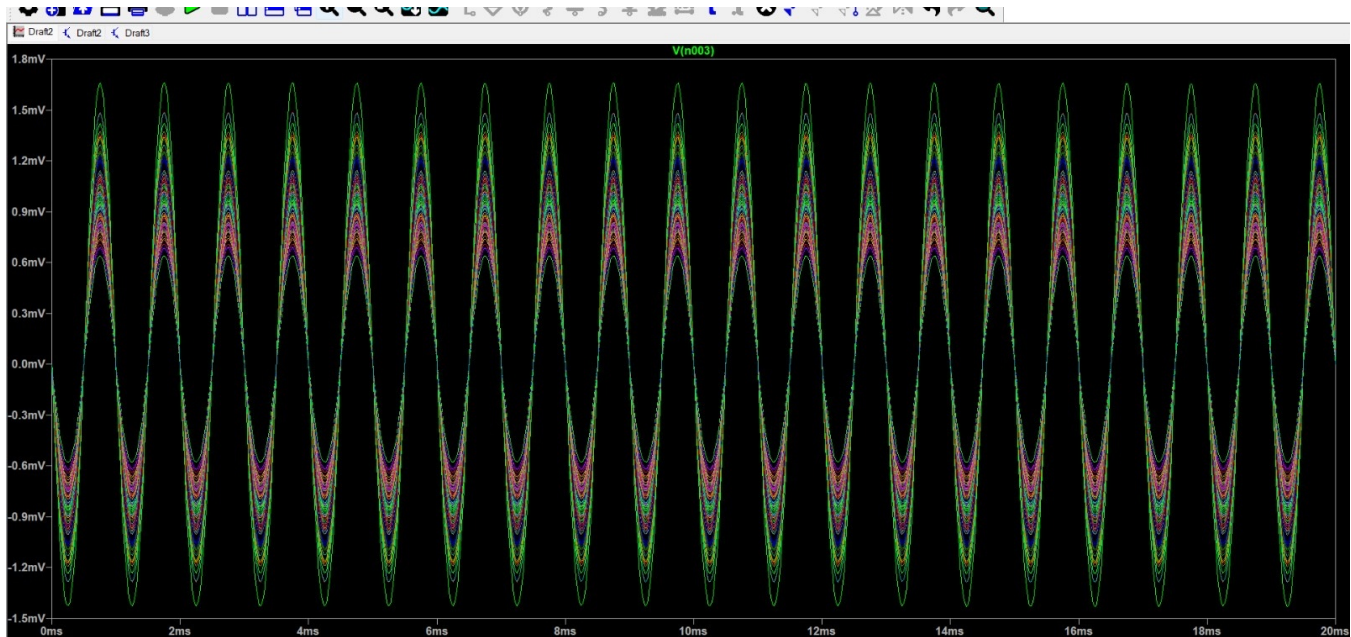


Figure 6. Monte Carlo output-voltage overlay. Each trace represents one randomized resistor-tolerance realization.

The overlay demonstrates output sensitivity to resistor variation. The waveform family remains bounded and periodic, indicating that the circuit does not collapse, rail, or become unstable under the modeled passive tolerance range.

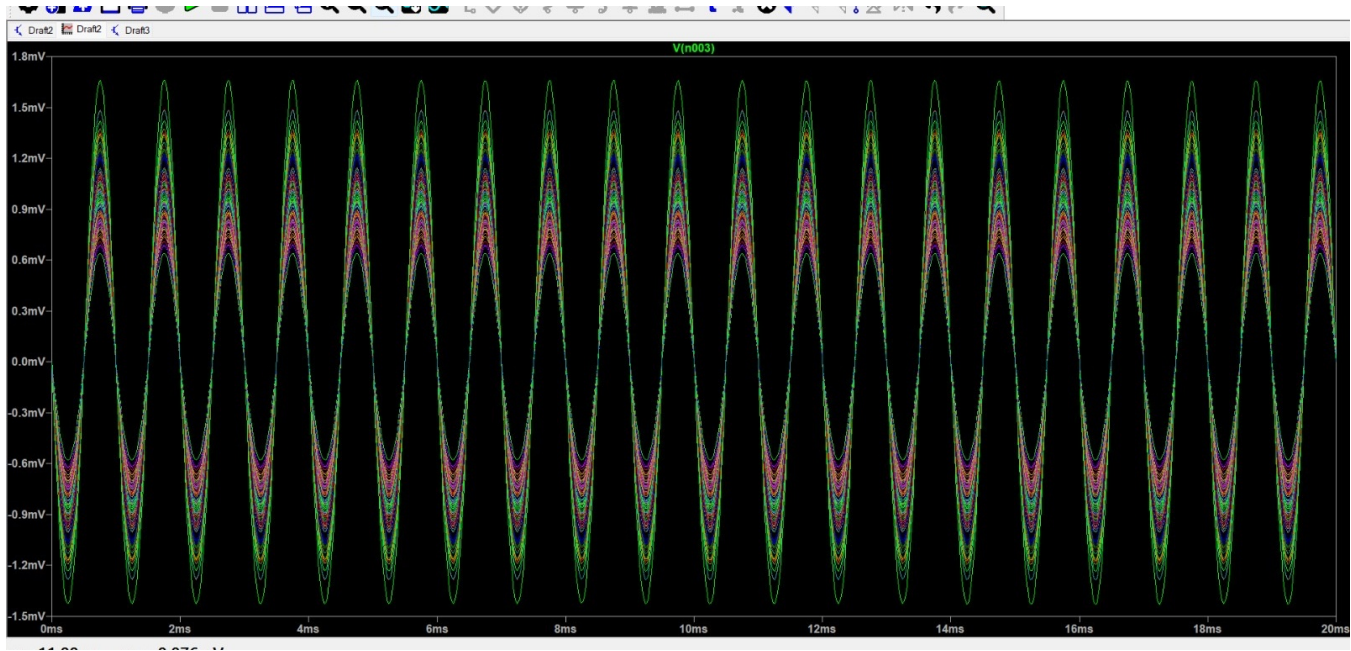


Figure 7. Zoomed Monte Carlo waveform envelope. The spread of the traces visually represents gain/output-amplitude sensitivity.

7. Bias and Current Variation Under Monte Carlo

The Monte Carlo drain-bias plot shows that the DC drain operating point shifts significantly across tolerance combinations. This is expected because the gate divider, source resistor, and drain resistor all affect the transistor current and therefore the drain voltage.

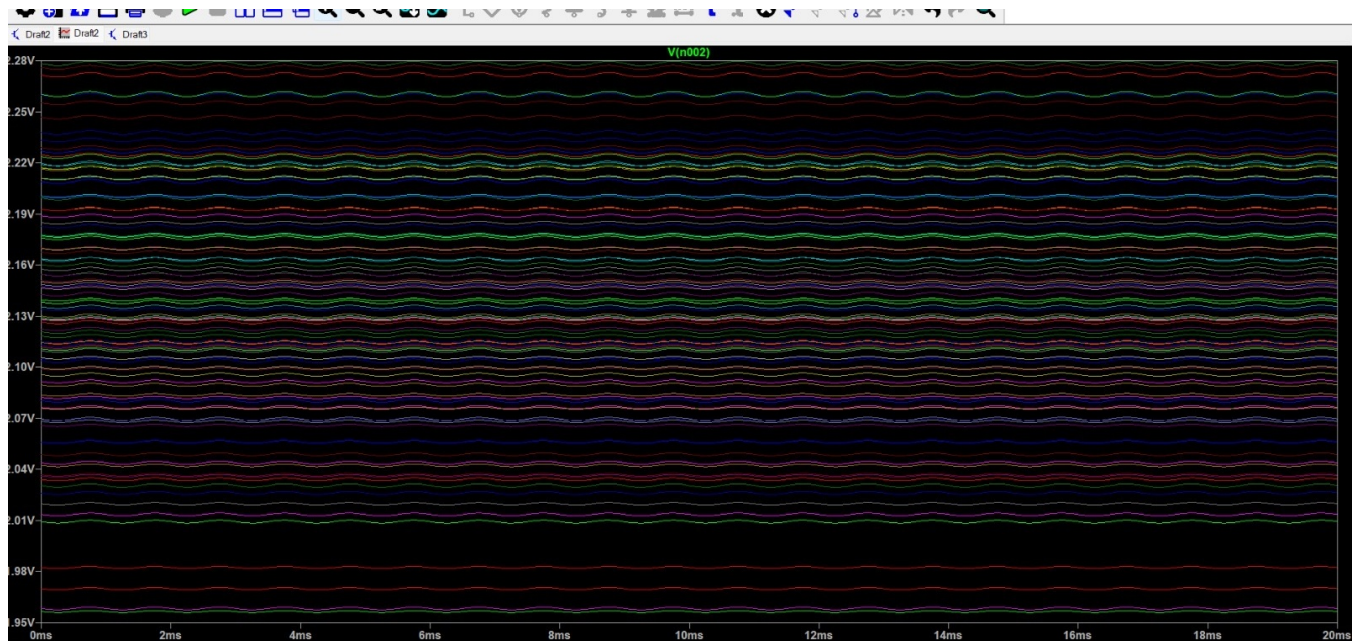


Figure 8. Monte Carlo drain-bias variation across 100 tolerance cases. The plot shows the spread in DC operating point caused by resistor tolerance variation.

The uploaded current plot confirms the same behavior from the current-domain perspective: different randomized resistor values shift drain current. Since the common-source stage converts current variation into drain-voltage variation through R_D , current spread directly translates into bias spread and small-signal response changes.

Final Simulation View

The figure below combines the schematic and representative Monte Carlo waveform view into one final project snapshot. It is suitable as a repository cover image or portfolio preview.

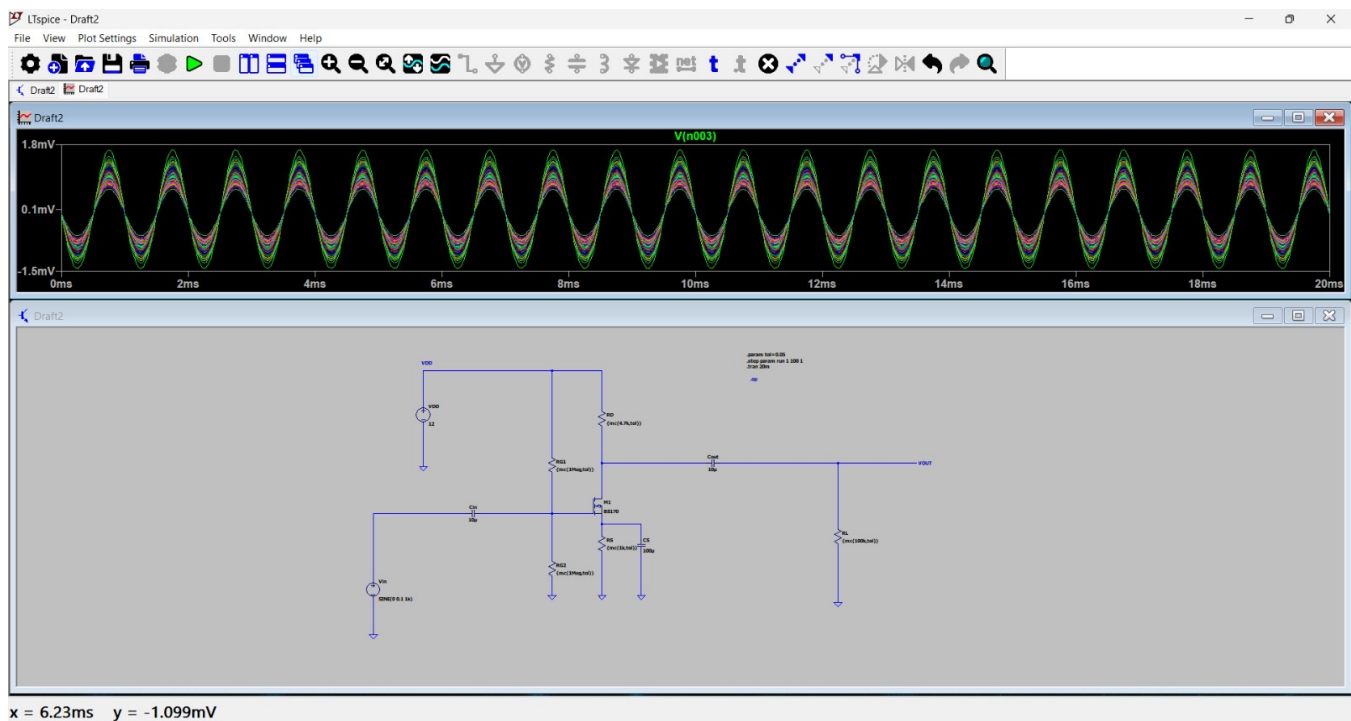


Figure 9. Final schematic and representative Monte Carlo waveform view. This figure summarizes the completed circuit and its statistical waveform behavior.

8. Simulation Debugging and Model-Control Notes

A useful part of the project workflow was debugging the dependency between Monte Carlo resistor expressions and the tolerance parameter. The LTspice error log below shows a condition where resistor values using mc(...,tol) were evaluated without the tolerance parameter being available to that simulation run. The correction is to keep the .param tol statement active whenever mc(...,tol) expressions remain in component values. For nominal single-run plots, tol can be set to 0 while preserving the same schematic.

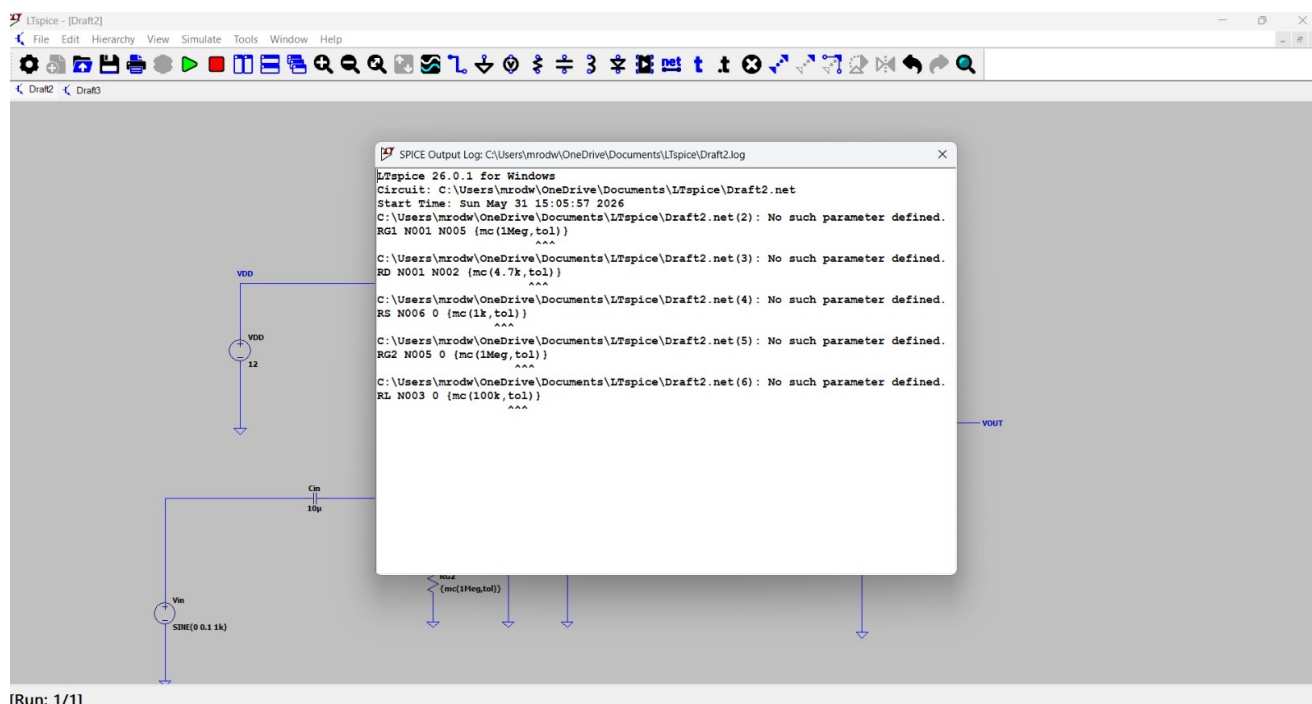


Figure 10. LTspice error-log screenshot from the debugging stage. The issue was resolved by defining tol before using mc(...,tol) component values.

Correct directive patterns used in the final workflow:

Normal single-run plot with Monte Carlo values still present:
 .param tol=0
 .tran 20m

Monte Carlo plot:
 .param tol=0.05
 .step param run 1 100 1
 .tran 20m

Operating-point inspection:
 .param tol=0
 .op

9. Results Summary

Metric / observation	Result	Interpretation
Input stimulus	100 mV amplitude, 1 kHz	Small-signal AC test condition
Nominal gate bias	6.0 V calculated	Set by equal 1 MOhm divider
Gate-bias tolerance range	Approx. 5.70 V to 6.30 V	+/-5% bias-divider tolerance sensitivity
Nominal drain voltage	Approx. 2.14 V from waveform	Transistor conducting strongly; drain is below mid-supply
Estimated RD current	Approx. 2.10 mA	Consistent with $(12 - 2.14)/4.7k$
Monte Carlo output spread	Bounded waveform envelope	Resistor tolerances affect amplitude but preserve stable operation
Bias spread	Approx. 1.95 V to 2.28 V in shown DCO	Operating point moves with passive variation
Design learning	Bias should be optimized for gain	Future revision can center drain near mid-supply

The project demonstrates a realistic analog verification mindset. The simulations do not simply prove that one waveform exists; they reveal how bias, current, and passive tolerances interact. The Monte Carlo results are especially valuable because they show the circuit as a distribution of possible manufactured outcomes rather than a single idealized case.

10. Skills Demonstrated

Skill area	Evidence in project
Analog circuit design	Common-source MOSFET amplifier with biasing, coupling, drain load, and source degeneration.
Device-level simulation	BS170 MOSFET model used in LTspice with transistor current and node-voltage probing.
Bias analysis	Gate divider calculation, drain-current estimate, and interpretation of drain operating point.
Signal analysis	Input, drain, output, and current waveforms interpreted in time domain.
Tolerance modeling	+/-5% resistor variation implemented using Monte Carlo expressions.
Statistical verification	100-run waveform overlays used to evaluate robustness and sensitivity.
Troubleshooting	Resolved parameter-definition issue with mc(...,tol) expressions and simulation directives.
Technical documentation	Report-style explanation tying schematic, calculations, waveforms, and verification results together.

Repository README summary:

Designed and simulated a common-source BS170 NMOS amplifier in LTspice. The project includes bias-network calculations, transient waveform analysis, drain-current interpretation, and 100-run Monte Carlo tolerance analysis using +/-5% resistor variation. Results demonstrate amplifier operation, passive-tolerance sensitivity, and a practical analog verification workflow.

Resume bullet:

Designed and verified a common-source NMOS amplifier in LTspice, implementing bias calculations, transient waveform analysis, drain-current probing, and 100-run Monte Carlo tolerance simulations to evaluate robustness under component variation.

11. Conclusion

This project completed a full simulation-based analog verification workflow for a common-source NMOS amplifier. The design was constructed in LTspice, driven by a defined sinusoidal input source, evaluated with transient analysis, and extended using Monte Carlo tolerance simulation. The results show how resistor tolerance affects output waveform amplitude, drain bias, and drain current.

The most important engineering result is not just the waveform itself, but the ability to reason from circuit topology to bias point, from bias point to current, and from current variation to tolerance-induced output variation. The project also identifies a clear next design improvement: re-biasing the transistor so the drain sits closer to mid-supply, which would increase available voltage swing and improve voltage-gain behavior.

Overall, this work demonstrates analog circuit simulation, tolerance-aware design thinking, LTspice proficiency, and technical documentation discipline in a portfolio-ready format.

Appendix: Copyable Final LTspice Directives

Monte Carlo mode:

```
.param tol=0.05  
.step param run 1 100 1  
.tran 20m
```

Nominal mode with same schematic values:

```
.param tol=0  
.tran 20m
```

Operating-point mode:

```
.param tol=0  
.op
```